

Comparative analysis of testicular histology in eight species of frogs.

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Anurans testes are paired and dorsally located near of kidneys. This organ is composed by seminiferous and connective epithelium varying between species. We aimed to compare testes morphology in eight species, *Physalaemus centralis*, *P. cuvieri*, *P. marmoratus*, *P. olfersii*, *Pseudopaludicola saltica* (Leiuperinae) and *Crossodactylus caramaschi*, *Hylodes magalhaesi* and *H. sazimai* (Hylodidae). We used morphometric analyzes (area: locular, interstitial, collagen and bundle spermatozoon; length: locular diameter and reticular fibers beam). We used five specimens of each species collected in reproductive period in southeast of Sao Paulo state, Brazil. Testes were removed and submitted to histological routine to embedding paraffin, historesin and stained with Hematoxylin/Eosin, Gomori's Reticulin and Picrossirius, images were captured and analyzed by Image-Pro Plus version 6.0. Analysis in stereoscopic microscope and two-dimensional reconstruction showed that the morphology of the seminiferous locules differ among the groups, being flattened in Hylodidae and spherical in Leiuperinae. Consequently, the Hylodes showed higher locules diameters being the smallest diameter in this group of *H. sazimai* having value next to *P. olfersii*, that have higher locular diameter of Leiuperinae. The locular area is larger in *H. cardosoi* and *H. sazimai*, being that this has the smaller interstitial areas. The Leiuperinae showed smaller areas of collagen except *P. marmoratus* showing largest area. However, this subfamily had higher thickness of reticular fibers beam except for *H. cardosoi* who presented fourth largest value. *P. olfersii* and *P. marmoratus* were the species with the largest areas of bundle spermatozoon featuring three times higher values than other species. Thus, it was observed that each species has unique characteristics that vary among these testicular components and some have values the morphological characteristics similar between species of the same group.

Keywords: Testes, Anuros, Morphology, Leiuperinae, Hylodidae.